

Yasemin Ozkut

MS Candidate | Computer Vision Engineer Intern | AI/ML/Computer Vision New Grad (May 2026)

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Education

MS	The Ohio State University (OSU) , Electrical and Computer Engineering • Focus: Computer Vision, Machine Learning, Multimodal Learning, VLMs	Columbus, OH, USA Aug 2024 – Expected May 2026
BS	Sabanci University , Computer Science and Engineering • Minor: Business Analytics, School of Management	Istanbul, Turkey Sept 2019 – Feb 2024

Skills

Languages: Python (3+ years hands-on, 1+ year professional), C++, SQL, JavaScript

Frameworks & Libraries: PyTorch, Pytorch Lightning, CUDA, YOLO, Keras-OCR, OpenCV, Unsloth, Hugging Face, Monai, Hydra, LangChain, LangGraph, Pandas, NumPy, Scikit-learn, React

Techniques: Machine Learning, Deep Learning, Computer Vision, Multimodal Vision-Language Models (VLMs/LLMs), Distributed Training, Prompt Engineering, AI-Assisted Coding, AI Agents, Fine-Tuning, Hyperparameter Tuning

Tools: Docker, Flask, Git, SLURM, Linux, Reachy Mini SDK

Publications

ERDES: A Benchmark Video Dataset for Retinal Detachment and Macular Status Classification in Ocular Ultrasound , <i>Nature Scientific Data</i> 🔗 Website Hugging Face Yasemin Ozkut , Pouyan Navard, Sirikar Adhikari, Alper Yilmaz	In revisions July 2025
HOPE: A Benchmark Dataset for Pulmonary Embolism Classification in Spatiotemporal Heart Ultrasound , <i>Nature Scientific Data</i> Yasemin Ozkut , Sirikar Adhikari, Alper Yilmaz	Under preparation Sept 2025 - Present

Experience

Ubihere Computer Vision Engineer Intern (Full-Time) • Wrapped research-stage person re-identification modules into a GPU-accelerated Flask microservice with REST API, now integrated into core product. • Reduced inference latency by 67% through GPU batching and parallelized image I/O. • Containerized with Docker and NVIDIA Container Toolkit, implementing health checks, logging, and resource controls for production deployment. • Currently developing an AI-powered admin dashboard that adapts to customer camera environments and delivers spatially aware tracking insights integrated with floor-plan data.	Aug 2025 – Present Columbus, OH
Photogrammetric Computer Vision Lab (PCVLab) 🐙 Graduate Research Assistant – Applied ML/Computer Vision • Developed two-stage diagnostic pipeline for retinal detachment (0.94–0.99 AUC) and macular status (0.73–0.88 AUC) classification in ocular ultrasound. • Benchmarked 8 CNN/ViT architectures using Lightning-Hydra, released dataset with 6K+ downloads. • Fine-tuned ViT + LLaMA 3.2 (via Unsloth) on 100K+ MIMIC-CXR chest X-ray images for automated report generation. • Scaled multi-GPU training on HPC with Hugging Face Accelerate and SLURM.	Aug 2024 – Aug 2025 Columbus, OH
DAI-Labor 🐙 AI/ML Researcher Intern (supported by Erasmus+ mobility program) • Annotated 1.2K product images, expanded to 3.5K with augmentations using Roboflow. • Trained YOLOv8 on German supermarket products (87.2% accuracy) and built a real-time detection pipeline. • Integrated Keras-OCR for German-language labels and benchmarked against YOLOv8 detection.	July 2023 – Sept 2023 Berlin, Germany

Projects

Judgy Reachy – Reachy Mini Robotics App, Personal Project | [Hugging Face](#)

January 2026

- Built a production-ready Reachy Mini robot (Hugging Face x Pollen Robotics) app that detects phone usage in real-time and delivers feedback via robot motion and speech.
- Developed computer vision pipeline using YOLO26n with cross-platform GPU acceleration (CUDA/MPS/CPU).
- Integrated Llama 3.1-8B via Groq API with 9 personality profiles, ElevenLabs API for natural text-to-speech, and expressive robot motion (head, antenna, body).
- Published as an open-source Hugging Face Community App, featured in the Reachy Mini Space ecosystem, supporting both simulation and physical robot deployment.

Agentic Person Re-Identification (ReID) via LangGraph and Vision-Language Reasoning,

July 2025

Personal Project 

- Built a LangGraph pipeline integrating YOLOv8, Qwen2.5-VL-3B, and Qwen2.5-7B for person detection, description generation, and identity assignment across frames/videos.
- Designed structured JSON prompts to extract rich semantic attributes (hair, clothing, accessories, face shape) and implemented persistent memory for global ID tracking.
- Enabled explainable matching with logged reasoning, confidence scores, and memory updates for re-identification.

Serious Game for Children with Cerebral Palsy, Sabanci University

Feb 2023 – Feb 2024

- Developed a serious game in Unity using C# for balance physiotherapy sessions in children with Cerebral Palsy.
- Collected live data from Wii Balance Board for each game session and stored it into MongoDB.
- Generated visual reports from the processed data using Python and Flask.

Spotify Music Artist Success Collaboration Network, Sabanci University

Feb 2023 – June 2023

- Collected and preprocessed Spotify and Kaggle data to build a collaboration network of 8.6K artists and 13.3K connections using Python/NetworkX, incorporating genre, popularity, and centrality metrics.
- Applied clustering and centrality algorithms to identify key influencers, and visualized the network with Gephi for success analysis.

Activities & Interests

Hiphop Dancer (2015–Present); **Board Member**, SuDance Club, Sabanci University (2019–2023); **Contestant**, Peak Games Unithon (2022); **Tutor**, Civic Involvement Project – 4th-grade students (2020); **Winter Sports Enthusiast** (Skiing, Snowboarding, Ice Skating)